File

Name: P3-K01-spur
Description: KISSsoft example
Changed by: AAA on: 30.10.2026 at: 19:35:06

Micropitting according to ISO/TS 6336-22

1 Overview

Lubricant load according to FVA Info sheet 54/7 10, Klüberoil GEM 1-220 N with details about wear coefficient kw
Reference data FZG-C Test:

Torque (Nm)	[T1Ref]	265.100
Line load at contact point A (N/mm)	[FbbRef,A]	236.300
Oil temperature (°C)	[θOilRef]	90.000
Tooth mass temperature (°C)	[θMRef]	127.666
Contact temperature (°C)	[θBRef,A]	251.735
Lubrication gap thickness (µm)	[hRef,A]	0.051
Specific film thickness in test	[λGFT]	0.102

1.1 Calculated for gear pair 1-2

Calculation at point (0:A, 1:AB, 2:B, 3:C, 4:D, 5:DE, 6:E, -1:No Point)			0	
Diameter (mm)	[dy]	143.447		465.018
Relative radius of curvature (mm)	[ρred]		11.603	
Load sharing factor	[XY]		0.333	
Contact stress (N/mm²)	[pH]		736.406	
Contact stress (N/mm²)	[pdyn]		855.569	
Minimal specific film thickness	[λGFY]		0.238	(h_Y=0.143 µm)
Safety against micropitting	[Sλ(B)]		1.667	

2 Calculation with nominal load

2.1 Calculated for gear pair 1-2

Equivalent Young's modulus (N/mm²)	[Er]	226373.6	
Roughness average value (µm)	[Ra]	0.600	
Elasticity coefficient (√N/mm)	[ZE]	189.812	
Lubricant factor	[XL]	1.000	
Lubrication coefficient for lubrication type	[XS]	1.000	
Tip relief factor	[XCa]	1.000	
Coefficient of friction	[µm]	0.061	
Tooth mass temperature (°C)	[θM]	84.456	
Dynamic viscosity (Ns/m²)	[ηtM]	25.720	
Pressure-viscosity coefficient (m²/N)	[α38]	0.0216	$\cdot 10^{-6}$
Pressure-viscosity coefficient at mass temperature (m²/N)	[αθM]	0.0170	$\cdot 10^{-6}$

	A	AB	B	C	D	DE	E
g (mm)	0.000	5.921	11.843	12.338	17.713	23.634	29.555
Γ	-0.481	-0.250	-0.019	0.000	0.210	0.440	0.671
d_1 (mm)	143.447	146.109	149.664	150.000	154.008	159.150	164.982
d_2 (mm)	465.018	460.547	456.340	456.000	452.436	448.775	445.399

$\rho_{t,1}$ (mm)	13.314	19.235	25.156	25.652	31.027	36.948	42.869
$\rho_{t,2}$ (mm)	90.318	84.397	78.476	77.981	72.606	66.684	60.763
ρ_n (mm)	11.603	15.665	19.050	19.302	21.737	23.775	25.136
X	0.333	0.500	1.000	1.000	1.000	0.500	0.333
p_H (N/mm²)	736.406	776.230	995.461	988.931	931.889	630.078	500.338
p_{dyn} (N/mm²)	855.569	901.837	1156.543	1148.957	1082.685	732.036	581.301
$v_{r,1}$ (m/s)	0.615	0.888	1.161	1.184	1.432	1.706	1.979
$v_{r,2}$ (m/s)	1.371	1.282	1.192	1.184	1.102	1.013	0.923
v_g (m/s)	-0.757	-0.394	-0.030	0.000	0.330	0.693	1.056
v_Σ (m/s)	1.986	2.169	2.353	2.368	2.535	2.718	2.902
θ_{fl} (Δ°C)	24.197	14.913	1.762	0.000	17.872	21.221	22.986
θ_B (°C)	108.652	99.369	86.218	84.456	102.328	105.676	107.442
G_M	3838.474	3838.474	3838.474	3838.474	3838.474	3838.474	3838.474
U	$9.723 \cdot 10^{-12}$	$7.867 \cdot 10^{-12}$	$7.016 \cdot 10^{-12}$	$6.970 \cdot 10^{-12}$	$6.624 \cdot 10^{-12}$	$6.495 \cdot 10^{-12}$	$6.558 \cdot 10^{-12}$
W	$0.898 \cdot 10^{-4}$	$0.997 \cdot 10^{-4}$	$1.640 \cdot 10^{-4}$	$1.619 \cdot 10^{-4}$	$1.437 \cdot 10^{-4}$	$0.657 \cdot 10^{-4}$	$0.414 \cdot 10^{-4}$
$\alpha_{\theta B}$ (m²/N)	$0.015 \cdot 10^{-6}$	$0.016 \cdot 10^{-6}$	$0.017 \cdot 10^{-6}$	$0.017 \cdot 10^{-6}$	$0.015 \cdot 10^{-6}$	$0.015 \cdot 10^{-6}$	$0.015 \cdot 10^{-6}$
S_{GF}	0.427	0.579	0.934	1.000	0.524	0.469	0.443
h (µm)	0.1430	0.1756	0.2053	0.2105	0.2015	0.2349	0.2622
λ_{GF}	0.238	0.293	0.342	0.351	0.336	0.392	0.437
S_λ	1.667	2.047	2.393	2.454	2.349	2.739	3.057

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